Digitalisering för bevarande och tillgänglighet
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SPADFABRIKEN KARLSTAD

Katalog å handredskap

A digitisation project by:
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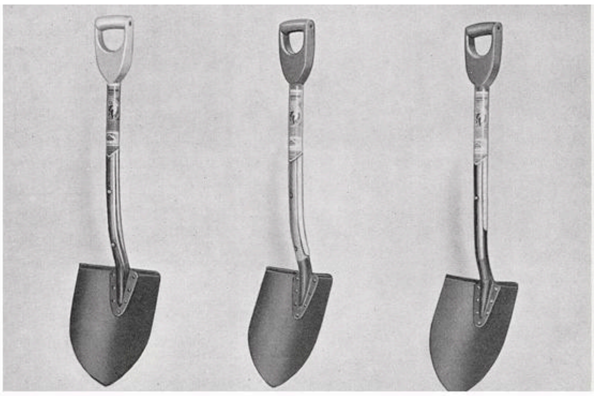

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1. Introduction

1.1. Cultural heritage preservation and ephemera

In this project we have digitised a catalogue displaying products from a company called Spadfabriken Karlstad. The catalogue was published in 1943 and contains a wide range of hand tools: spades, shovels, grabs, rakes and other gardening tools.

The catalogue belongs to the category of ephemera, which is publications with a short lifespan and temporary informational value. Our catalogue, for example, was published in 1943, but we can assume that the company later published an updated version of the catalogue, making the one we have digitised obsolete.

The word itself, ephemera, derives from the ancient Greek ἐφήμερος, which means "lasting only a day" (Logeion, n.d.). Even though the ephemera only "lasts one day" in one sense, information can be found in such documents that is of great value from an historical perspective. Ephemera reflects the time and context it was created in (Kungliga biblioteket, 2025). In our case, the catalogue shows standards of the 1940s such as pricing, marketing trends, linguistics and more.

The National Library of Sweden (Kungliga biblioteket) as well as the university libraries of Uppsala and Lund all have a vast collection of ephemera. The material is not in the library catalogue, but it is possible to make requests for publications and a small part of the big collection of ephemera has been digitized by the National Library of Sweden.

1.2. Background and context

The item digitised is a catalogue from Spadbariken Karlstad, a factory producing shovels and other hand tools. Originally the factory was founded in 1918 under a different name, and during the years the name and location of the company have changed several times. The shovel company was almost destroyed after a brutal fire in 1927, but they managed to get back on their feet and even develop the range of products. The catalogue from 1943 is from this era. Today, the company is called Karlstad Redskap and is still manufacturing tools.

The copy of the catalogue is owned by the Karlstad Local Heritage Association. The association works for the purpose of protecting and taking care of the local cultural heritage and they collect and digitise photos and documents. Karlstad Local Heritage Association publish their documents on a web page called wermlandsbilder.se, a homepage that is a collaborative project between a number of associations in Värmland.

1.4. Legal aspects

According to Swedish copyright law (SFS 1960:729), the copyright is free 70 years after the copyright holder has passed away. If the material, as in our case, doesn't have a named creator, the rights are considered free 70 years after the publication of the document.
We investigated if our catalogue had any kind of named author but as is often the case with ephemera there is no author behind the publication.

After doing a google image search on the illustration on the cover of the catalogue, we discovered that our old ephemera actually had connections to a modern company existing today, called Karlstad Redskap. We contacted the company and they granted us permission to use the catalogue in our digitising project. They didn't mind us publishing the document online. But as their logo is on the cover, we acted as if they were in fact copyright holders and arrived at the decision using following rights statements: *Copyright is held by Karlstad Redskap AB. The material has been digitised and published with permission from the copyright holder*.

2. Methodology

2.1. Selection of Material

When searching for material, we found the website Wermlandsbilder.se, which collects digitised material from local heritage associations in Värmland. We were particularly drawn to their digitised ephemera and contacted them regarding a possible collaboration and came into contact with Karlstad's local heritage association, who invited us to look through their archive for a suitable object.

Cowick (2018, pp. 11–12) presents several criteria to consider when selecting material for digitisation, including historical or research value, legal and ethical issues, physical condition, demand, and whether the material has already been digitised.

In our selection process, we mainly prioritised value, legal considerations and that it was not already digitised. We were interested in ephemera because such material is often overlooked and may be rare or unique. To avoid copyright issues, we focused on older material and objects without identified creators. We also wanted an item of manageable size due to limited time and resources, preferably containing both text and images. Based on these considerations, we selected a 1943 catalogue from the company Spadfabriken Karlstad.

2.2. Image Capture and Image Editing

2.2.1. Image Capture

Several methods for image capture were tested during the project. We experimented with photography at the University of Borås, but found it difficult to achieve even lighting, and the catalogue's semi-glossy paper surface caused reflections. We tested scanning the catalogue using a book scanner at the Karlstad heritage association. However, since the scanner was optimized for text capture and mass digitisation, we did not consider it suitable for our material.

Instead, the image capture was carried out using a flatbed scanner (Canon PIXMA TS3751i). Flatbed scanners involve direct contact with the material and are not suitable for delicate objects (DFG, 2013, p. 15), but our catalogue was in good condition and could be scanned safely.

Test scans were carried out using white, grey, and black 300 gsm paper backgrounds. A grey background was selected as it presented the material most clearly. The catalogue was scanned page by page at 300 dpi and saved as TIFF files. The method was effective, producing good colour reproduction without shadows or distortion near the gutter.

2.2.2. Image Editing and File Naming

JPEG derivatives of the master images were created for use on the homepage and were slightly edited, using Adobe Photoshop version 24.1.1, to represent the material as authentic as possible and leaving a small border of the grey background around the pages. Editing included cropping, straightening, and minor colour adjustments. According to the DFG guidelines (2013, p. 18), post-processing may be necessary to optimise image quality while avoiding manipulations that alter the object itself.

For file naming, we consulted Cowick's (2018, pp. 37–39) guidelines. Cowick argues that filenames should be clear, concise, and sufficiently descriptive to not lose their context if relocated. Special characters should be avoided since they may cause errors in digital environments, hyphens are recommended. Cowick also recommends preceding single-digit numbers with a zero.

Based on these guidelines, the master files were named according to the format *Spadfabriken-Karlstad-katalog-1943-01-master.tif*, while the JPEG derivatives followed the format *Spadfabriken-Karlstad-katalog-1943-01.jpg*.

2.3. Transcription and Text Capture

Text capture can be done by manual transcription or by using an OCR reader to extract the text. According to Tanner (2004, p.4) there are a number of things to consider when you are considering using OCR or not. The main reasons to use OCR is accuracy and efficiency.

One OCR-test was however carried out to investigate the process and if it was desirable in our project. We used Google docs which is an easy, accessible tool for OCR-reading. The test was made on one of the jpeg-files containing numbers and a small amount of text apart from illustrations and the results were very good.

However, the catalogue didn't contain a lot of text and even if the accuracy was high in our OCR-test, the manual transcription was still more time efficient and we did the rest of the transcription manually. The transcription was proofread by the other group members to ensure that there were no errors.

2.4. Text Encoding

The text was encoded using the Text Encoding Initiative (TEI) standard P5 Version. TEI is described in DFG (2013, p. 33) as "[…] prospectively the best choice for long-term archiving as long as it is carefully documented".

TEI is a flexible way to encode text, where the encoder can choose the level of complexity to suit the project (Renear, 2004). It also offers a way to store metadata about the project, information about images, and contextual information.

While working on the encoding, the TEI Guidelines was consulted as well as TEI by Example, TEI Lite and workshop-material from the University of Borås.

The TEI header stores the project metadata. More about this in section 2.5.2. We included a <facsimile>-section which lists all images, their location in GitHub and their descriptions.

In the <text>-section we used divs and page breaks to show where a new page started in the material. We chose to tag much of the text with <hi rend> to show different layout features in the text such as bold, centred and italic etc. We chose to use <ab> (anonymous block) instead of using <p> when encoding the article numbers of the products. The anonymous block "contains any component-level unit of text, acting as a container for phrase or inter level elements analogous to, but without the same constraints as, a paragraph" (TEI Consortium, 2026). This approach was considered suitable since the article numbers are more stand alone units of information than the rest of the text.

Lastly we chose to incorporate information from the price list under the tag <standOff>, which can be used to store contextual information.

We chose to use Swedish as the main language in the TEI code since we mainly foresee a Swedish audience for the material. With the exception of the copyright statement which is also in English.

2.5. Metadata

Metadata is an important part of digitisation. It makes objects searchable and provides contextual information (DFG, 2013, p. 26). We added metadata in the TEI header and embedded in the image files. The details and considerations surrounding these implementations are described in the coming subsections.

The DFG (2013, p. 26) emphasises the importance of using standard-compliant and software-neutral metadata formats, commonly based on XML encoding. Since TEI is XML-based and follows established standards, our use aligns with these recommendations.

The subject terms used in the metadata were assigned using the controlled vocabulary Svenska ämnesord. More specific terms were prioritised over broader alternatives in accordance with Dublin Core guidelines (Dublin Core Metadata Initiative, n.d.). For example, the term "trädgårdsredskap" instead of the general term "redskap".

In addition to terms describing the material type and content, such as "kataloger" and "lantbruksredskap", the term "reklam" was included to reflect its commercial function. The agricultural vocabulary AGROVOC was considered for more granular terminology, but was not implemented due to the Swedish-language focus of the project.

2.5.1. Embedded Metadata

Embedded metadata was added to the TIFF and JPEG files using Adobe Photoshop version 24.1.1. The metadata fields available in Photoshop are presented using names corresponding to Dublin Core elements, and Dublin Core definitions were therefore consulted when determining what information to include (Dublin Core Metadata Initiative, n.d.).

Some relevant Dublin Core elements such as creator, type, and date, did not have corresponding fields in Photoshop. In these cases, the information was incorporated into the Description field.

The embedded metadata also included information about the objects shown on each catalogue page. Combined expressions such as “gräv- och dikesspadar” were rewritten as separate terms “grävspadar” and “dikesspadar” to improve clarity and searchability.

2.5.2 TEI-header metadata

The TEI header offers a way to describe metadata about the project and we included both mandatory and optional tags that we found valuable for the project.

In <fileDesc> we included: <titleStmt> title of the project, original author, creators of the TEI-file etc., <publicationStmt> copyright declaration in both Swedish and English etc., <sourceDesc> for example information about the source and also a physical description of the printed matter.

We also included: <profileDesc> where we stated keywords with URI:s from Svenska ämnesord, <encodingDesc> where information about the project and encoding choices are stated and <xenoData> which contains metadata from one of the scanned images.

2.6. Digital Presentation

The result of the project is presented on a webpage published through GitHub Pages using a template provided through the course. The webpage includes links to files stored in the project’s GitHub repository. The aim of the website was to present the result of the digitisation project and to provide contextual information about the catalogue.

The homepage consists of a landing page featuring a collage of images intended to create interest and a short introduction to the content of the website. Two different ways of viewing the material are then provided. The page “Transkribering” presents each catalogue page from top to bottom alongside the transcribed text displayed next to the images.

On the webpage the transcription is presented as left-aligned text and the numbers are shown as a list, regardless of the original layout. Readability was prioritised over layout accuracy because we concluded that the work needed to reproduce the layout was considered disproportionate to its value for the project and users. The images are presented next to the transcription where one can see the layout.

The page “Information från prislista” only includes the pages containing products. The accompanying text consists of information from a separate price list previously digitised by the local heritage association. The page “Historia” presents historical information about the company and an image from the archive of Värmlands Museum to provide additional context. The final page, “Om projektet”, provides information about the project itself and links to high-resolution image files and the TEI file stored in the project’s GitHub repository.

A PDF version of the material will be delivered to the local heritage association and may later become available through Wermlandsbilder.se.

3. Long-term Preservation

Digitisation is not the same as long-term preservation. As Cowick (2018, p. 3) points out, digitisation primarily concerns creating digital access, while digital preservation involves ensuring that the material remains accessible and usable over time. Long-term preservation requires both suitable file formats and secure storage solutions.

The DFG (2013, p. 6) recommends TIFF as a format for digital master files. In accordance with these recommendations, the unedited master images created were saved as TIFF files. Edited JPEG derivatives were created for web presentation and access.

Cowick (2018, pp. 6, 32) recommends preserving multiple copies of digital files on different media and argues that preserving the digital object alone is not sufficient. The metadata must also be preserved.

Both the image files with embedded metadata and the TEI files are stored in the project GitHub repository, in a shared Google Drive folder, and locally on a storage device belonging to one of the group members. In addition, a PDF version of the material will be delivered to the local heritage association for further preservation and access.

4. Time Spent

During the course of the project, we have documented the hours spent on various activities in a shared journal. The total amount of time spent on the project is approximately 387 hours.

Below is a summary of the hours spent working on the project, divided into general categories.

Activity	Hours (total)
Group meetings and communication	112
Image capture and Image editing	18
Text encoding and Transcription	56
Research	22
Report	104
Metadata	29
Working with GitHub	12
Practical work around item (selection, collection)	12
Publishing and working with the webpage (HTML, CSS)	22
Totalt:	**387**

It is hard to estimate what a continuation of the project would require in time. If we were to increase the scope of the project, it all depends on what kind additional work that will be done. One could for example add another catalogue, and then it would be possible to benefit from our previous work and save time in the digitsing process.

5. Analysis and Discussion

5.1. Critical or Mass Digitisation

The concepts of mass digitisation and critical digitisation can be used to describe different approaches to digitisation. Dahlström (2011, p. 94) describes mass digitisation as a process where large quantities of material are digitised through automated workflows, minimising time-consuming manual labour and individual treatment of documents.

By contrast, Dahlström (2011, p. 97) describes critical digitisation as involving tailored decisions focused on the unique characteristics of individual documents. Some of the features associated with critical digitisation include deliberate selection of source material, descriptive metadata, text encoding, and attempts to produce representations that remain faithful to the visual and graphical qualities of the source.

Initially we categorised our project as a critical digitisation and this project may be seen as meeting some of the features associated with such an approach. High-resolution files were created, and efforts were made to reproduce the material as faithfully as possible. Metadata was created, text encoding was carried out, and the catalogues text was manually transcribed and proofread.

Dahlström (2011, p. 94) points out that contextual descriptions and similar labour-intensive work are generally not associated with mass digitisation. We conducted some research into the history of the company and incorporated additional material to provide further context. During the project, a previously digitised price list was identified. Information about specific product models was transcribed from the price list and linked to corresponding entries on the project website, adding additional contextual information.

However, there are also aspects of the project that reflect priorities more commonly associated with mass digitisation. The material was selected primarily based on interest and availability. Although contextual material was incorporated where possible, the broader context of the catalogue remains limited. Some choices were also made throughout the project with regard to efficiency and time constraints, and we were not able to explore the material as deeply or in as much detail as we could have or would have liked.

This reflects Dahlström's (2011, p. 99) argument that mass digitisation and critical digitisation should not be understood as strictly separated categories, but rather as positions along a continuum where projects combine features from both approaches.. Rather than fitting entirely within either a mass digitisation or a critical digitisation model, this project can therefore be understood as occupying a position between the two approaches.

5.2. Interpretation and Information loss in Digitisation

Björk (2015, p. 40) describes digitisation as an act of interpretation and points out that alterations are always introduced in the process. We made decisions with the intention of staying true to the material, but we are aware that much more could have been conveyed.

According to Björk (2015, p. 23), a digital reproduction is shaped by the goals of the digitisation project. It presents a specific perspective rather than an exact copy and may marginalise aspects that do not align with the project's central focus. Terras (2015, pp. 744–746) also argues that making digitised cultural heritage material openly accessible can create new opportunities for research, reuse, and analysis beyond the original purpose of the digitisation project.

This project was carried out from our perspective as Library and Information Science students, with inspiration from the National Library of Sweden's work with similar material. This is a specific perspective, and our digitisation focuses on certain aspects of the material while excluding others that may be important from a different point of view. If we speculate, aspects such as paper weight, the type of staples used, or the chemical composition of the sticker glue may be details that were excluded but could potentially be relevant in another context or field of research.

There are certainly aspects that we did not capture. However, we made sure not to intentionally exclude any information visible in the material during the digitisation process. For example preserving stains and creases in the paper. We made an effort to reproduce the colours authentically. On one page in the catalogue, a sticker is attached that creates shadows and creases on both sides of the page during scanning. We chose not to edit these out because they showed that something had been physically added to the page.

As an example of the losses involved in digitisation, Björk (2015, pp. 21–22) discusses the digitisation of the Codex Gigas. He argues that the imposing physical presence and size of the book, the smell of the parchment, and the sounds created while handling it are lost in the digital version. He also mentions that the physical presence of the book is most clearly represented in photographs from the digitisation process.

Although our material is much more modest in scale and character, there are still similar kinds of losses in our project. The feel and smell of the paper, or the sense that the catalogue seemed almost untouched, as if we were among the first people to open this copy, are not conveyed. With more forethought, we could, as in the example of the Codex Gigas, have documented the handling of the object during the digitisation process through photographs and presented these on the website. This could have provided a more complete understanding of both the object itself and the digitisation process.

Milekic (2007, pp. 369–388) also discusses the loss of information in digitisation and argues that digital tools can make virtual information more tangible, for example in digital archives of cultural heritage material. He gives examples such as zooming tools and digital magnifying functions that allow users to examine details more closely.

Since the catalogue contains many detailed illustrations, we also discussed implementing a zooming function. However, due to limitations in time and technical knowledge, this was not possible within the scope of the project. As a compromise, the images can be opened in a larger format in a separate window, allowing some details to be viewed more clearly.

We considered the option of using the TEI-tag <zone> to define the coordinates of the article numbers (and the product images), which may have been useful in a presentational feature on the web page. But since this is a small scale project we decided it was out of scope. Also the products all have unique article numbers that can be used to correlate any additional information to the different products. As on the “Information från prislista” page.

Furthermore we chose to present the catalogue as separate pages from top to bottom, alongside the transcription. As a result, the layout of the spreads, the experience of browsing through the catalogue, and the sense of it as a unified physical object became less apparent. Given more time, a digital implementation such as an interactive page-turning function could potentially provide a closer approximation of the original artefact.

6. Final Reflections

This project has provided valuable insight into how a digitisation project is carried out in practice. Throughout the project, we realised how many decisions and considerations are involved in digitisation and increased our understanding of the importance of preserving and making cultural heritage material accessible. Furthermore, we learned that there are more kinds of shovels than we ever could have imagined.

One challenge was determining where to draw the line regarding the scope of the project. It is almost always possible to add more or dig deeper. If more time and resources had been available, we would have liked to expand the project by finding and digitising additional catalogues, incorporating more contextual material, and developing more advanced functions for the website.

We would like to thank the Karlstad Local Heritage Association for giving us access to the material, Karlstad Redskap for granting us permission to use it, and our teachers at the University of Borås for their support and guidance throughout the project.

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